

Alejandro Santin

E-Mail: AlejandroSantin26@gmail.com

07828345754

Education:

University of Nottingham | Nottingham, England 08/2024-N/A

Studying for a Computer Science degree

Finchley Catholic High School | London, England 07/2017 – 05/2024

A Level

07/2022 – 05/2024

Subjects:

Computer Science -A | Maths-A | Physics-A

GCSE

07/ 2017 – 05/2022

Subjects:

Maths-9 | Computer Science-9 | Spanish-9

Physics-8 | Biology-8 | Chemistry-8

English Literature-7 | English Language-7 | French -7

Religious Education-7 | Design and Technology-7

Projects:

Software Maintenance & Evolution (Java)

Refactored a legacy Java Swing application into a modern JavaFX architecture using SOLID principles. Implemented automated testing with JUnit and Maven, while managing version control and CI/CD pipelines through GitLab.

Simulated OS (C)

Engineered a multi-threaded simulator using POSIX threads to manage process scheduling and synchronisation. Developed thread-safe priority queues and a round-robin scheduler with dynamic priority boosting in a Linux environment.

Data Search Website (HTML, CSS & JavaScript):

Developed a website frontend using HTML, CSS, and JavaScript, and integrated it with a backend database, enhancing understanding of frontend-backend communication and overall website functionality.

Encryption / Decryption program (Haskell):

Created an encryption/decryption program in Haskell using functional programming principles, gaining strong proficiency in Haskell and deepening functional programming skills.

OOP Game (Java):

Developed a two-player game with a graphical user interface (GUI) using object-oriented programming, including a basic AI for strategic moves. Strengthened OOP skills and gained valuable experience in comprehensive testing practices.

AI Pattern Recognition Software (Python):

Developed an AI in Python using imported libraries to analyse spreadsheet data, recognize patterns, and make trend-based predictions. Improved skills in data processing and machine learning concepts.

ALU & ARM decode project:

Built a 6502 ALU to better understand logic gates and CPU operations. Also developed an ARM decoder that translates binary instructions into plain English using the ARM7TDMI manual.

Client & Server Project (C):

Used Python to build a basic client-server system with successful network communication, strengthening understanding of networking principles and client-server interactions.

Mandelbrot in Assembly (Assembly):

Implemented assembly code to generate and display the Mandelbrot set, focusing on optimisation and accurate fractal rendering. This project deepened my understanding of low-level programming and the maths behind fractals.

A python maths quiz (Python):

Created a Python-based interactive maths quiz using OOP, hashing, linked lists, and SQL with encryption to support secure, dynamic, and scalable learning. Strengthened skills in secure coding and educational tool development.

Work Experience:

-
- I've volunteered at charities before. (11/2022-12/2022)
 - Additionally, I did waitering and worked behind the counter, both at my other uncle's restaurant for a short period of time. (06/2022-09/2022)
 - I currently work on a zero-hour contract as a lifeguard/ swim teacher where I learned leadership skills, problem solving skills and decision-making skills upon many others

Additional Information:

-
- Fluent in Spanish
 - Advanced pseudocode knowledge
 - NPQL certification

- AED certification
- SEQ Level 1 & 2
- Safeguarding qualified
- Junior lifeguard instructor qualified
- Mentor for first year student
- Assigned role of team leader for ongoing software engineering group project